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Carotid Sinus Stimulation for Hypertension and Heart Failure

Number: 0820

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Policy History

Last Review 03/03/2026

Effective: 09/20/2011

Next Review: 09/10/2026

Review History

Definitions

Policy

Scope of Policy

This Clinical Policy Bulletin addresses carotid sinus stimulation for the treatment of hypertension.

Experimental, Investigational, or Unproven

I. Aetna considers implantable carotid sinus stimulators (e.g., the Barostim neo™ System, and the Rheos Baroreflex Hypertension Therapy System) experimental, investigational, or unproven for the treatment of hypertension and for all other indications (e.g., heart failure) because its effectiveness has not been established.

II. Aetna considers magnetic stimulation of carotid sinus experimental, investigational, or unproven for the treatment of hypertension because its effectiveness has not been established.

Note: Removal of a device is covered when it is done for significant symptoms/issues (e.g., pain, infection); however, replacement of an investigational device is not covered.

Additional Information

Clinical Policy Bulletin Notes

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description

Code	Code Description
CPT codes not covered for indications listed in the CPB:	
0266T	Implantation or replacement of carotid sinus baroreflex activation device; total system (includes generator placement, unilateral or bilateral lead placement, intra-operative interrogation, programming and repositioning, when performed)
0267T	Implantation or replacement of carotid sinus baroreflex activation device; lead only (includes intra-operative interrogation, programming and repositioning, when performed)
0268T	Implantation or replacement of carotid sinus baroreflex activation device; pulse generator only (includes intra-operative interrogation, programming and repositioning, when performed)
0269T	Revision or removal of carotid sinus baroreflex activation device; total system (includes generator placement, unilateral or bilateral lead placement, intra-operative interrogation, programming and repositioning, when performed)
0270T	Revision or removal of carotid sinus baroreflex activation device; lead only (includes intra-operative interrogation, programming and repositioning, when performed)
0271T	Revision or removal of carotid sinus baroreflex activation device; pulse generator only (includes intra-operative interrogation, programming and repositioning, when performed)
0272T	Interrogation device evaluation (in person), carotid sinus baroreflex activation system, including telemetric iterative communications with the implantable device to monitor device diagnostics and programmed therapy values, with interpretation and report (eg, battery status, lead impedance, pulse amplitude, pulse width, therapy frequency, pathway mode, burst node, therapy start/stop times each day)

Code	Code Description
0273T	Interrogation device evaluation (in person), carotid sinus baroreflex activation system, including telemetric iterative communications with the implantable device to monitor device diagnostics and programmed therapy values, with interpretation and report (eg, battery status, lead impedance, pulse amplitude, pulse width, therapy frequency, pathway mode, burst node, therapy start/stop times each day) with programming
64654	Initial open implantation of baroreflex activation therapy (BAT) modulation system, including lead placement onto the carotid sinus, lead tunnelling, connection to a pulse generator placed in a distant subcutaneous pocket (ie, total system), and intraoperative interrogation and programming
64655	lead only
64656	Revision or replacement of baroreflex activation therapy (BAT) modulation system, with intraoperative interrogation and programming; pulse generator only
64657	Percutaneous electrical nerve field stimulation, cranial nerves, without implantation
64658	Removal of baroreflex activation therapy (BAT) modulation system; lead only
64659	pulse generator only
HCPCS codes not covered for indications listed in the CPB:	
C1825	Generator, neurostimulator (implantable), non-rechargeable with carotid sinus baroreceptor stimulation lead(s)
ICD-10 codes not covered for indications listed in the CPB (not all-inclusive):	
I10 - I16.9	Hypertensive diseases
I50.1 - I50.9	Heart failure

Background

Hypertension is a major cause of morbidity and mortality in the United States. Despite the availability of potent anti-hypertensive medications, many patients remain hypertensive. Thus, non-pharmacological therapies have been attracting more interest. Electrical stimulation (ES) of the carotid sinus has been shown to lowers blood pressure (BP) by activating the baroreflex and thereby reducing sympathetic tone. The arterial baroreflex regulates mean arterial pressure by responding automatically to changes in cardiac output and vessel tone via baroreceptors, which monitor arterial pressure by gauging the degree of stretch in vessel walls. Baroreceptors are located in the walls of the aorta and carotid arteries, with a concentration of receptors located within the carotid sinus. Activation of baroreceptors produces immediate responses in cardiovascular sympathetic and cardiac parasympathetic nerves to adjust heart rate (HR), stroke volume, vasoconstriction, as well as fluid excretion. These actions raise or lower BP as needed.

The Rheos Baroreflex Hypertension Therapy System (CVRx, Inc., Minneapolis, MN) is an implantable device for the treatment of patients with drug-resistant hypertension (i.e., the hypertensive state characterized by the inability of multiple anti-hypertensive drug interventions to lower BP to goal levels) who have a systolic BP (SBP) of greater than or equal to 160 mm Hg. It is reported to reduce BP by activating the baroreflex through ES of baroreceptors in the carotid sinus. The Rheos Baroreflex Hypertension Therapy System has 3 major components:

- An implantable pulse generator (IPG)
- Carotid sinus leads (CSLs)
- A programmer system.

The pulse generator is implanted subcutaneously near the collarbone. The CSLs are placed on the carotid arteries and run subcutaneously to the pulse generator (**Note**: the procedure does not involve carotid dissection). The operation usually takes about 2 to 3 hours. After implantation, a continuous mild electrical pulsation stimulates baroreceptors to produce a reflexive reduction in BP. The IPG provides control and delivery of activation energy through the CSLs. The leads conduct activation energy from the IPG to target receptors in the left and right carotid arteries. The programmer system provides the ability to communicate non-invasively with the IPG after implantation to modulate the frequency and intensity of ES.

In a phase II clinical trial, Illig et al (2006) evaluated the response of patients with multidrug-resistant hypertension to ES of the carotid sinus via an implantable device. The system consists of an IPG with bilateral peri-vascular CSLs. Implantation was performed bilaterally with patients under narcotic anesthesia. Dose-response testing at 0 to 6 V was assessed before discharge and at monthly intervals thereafter; the device was activated after 1 month's recovery time. This was a Food and Drug Administration-monitored phase II trial performed at 5 centers in the United States. A total of 10 patients with resistant hypertension (taking a median of 6 anti-hypertensive medications) underwent implantation. All 10 were successful, with no significant morbidity. The mean procedure time was 198 minutes. There were no adverse events attributable to the device. Pre-discharge dose-response testing revealed consistent ($r = 0.88$) reductions in SBP of 41 mm Hg (mean fall is from 180 to 139 mm Hg), with a peak response at 4.8 V ($p < 0.001$) and without significant bradycardia or bothersome symptoms. The authors concluded that a surgically implantable device for ES of the carotid baroreflex system can be placed safely and produces a significant acute decrease in BP without significant side effects.

Tordoir et al (2007) examined peri-operative outcomes and BP responses to an implantable carotid sinus baroreflex activating system being investigated for the treatment of drug-resistant hypertension. These investigators reported on the first 17 patients enrolled in a multi-center study. Bilateral peri-vascular CSLs and a IPG are permanently implanted. Optimal placement of the CSL is determined by intra-operative BP responses to test activations. Acute BP responses were tested post-operatively and during the first 4 months of follow-up. Prior to implant, BP was 189.6 +/- 27.5 (SBP)/110.7 +/- 15.3 (diastolic BP [DBP]) mm Hg despite stable therapy (5.2 +/- 1.8 anti-hypertensive drugs). The mean procedure time was 202 +/- 43 minutes. No peri-operative strokes or deaths occurred. System tests performed 1 or up to 3 days post-operatively resulted in significant (all p < or = 0.0001) mean maximum reduction, with standard deviations and 95 % confidence limits for SBP, DBP and HR of 28 +/- 22 (17, 39) mm Hg, 16 +/- 11 (10, 22) mm Hg and 8 +/- 4 (6, 11) beats/min, respectively. Repeated testing during 3 months of therapeutic electrical activation demonstrated a durable response. The authors concluded that these preliminary data suggest an acceptable safety of the procedure with a low rate of adverse events and support further clinical development of baroreflex activation as a new concept to treat resistant hypertension.

Heusser et al (2010) tested the hypothesis that ES of carotid baroreceptors acutely reduces sympathetic vasomotor tone and BP in patients with drug-resistant arterial hypertension. Furthermore, these researchers tested whether ES impairs the physiological baroreflex regulation. They studied 7 men and 5 women (aged 43 to 69 years) with drug-resistant arterial hypertension. A bilateral electric baroreflex stimulator at the level of the carotid sinus (Rheos) was implanted greater than or equal to 1 month before the study. Intra-arterial BP, HR, muscle sympathetic nerve activity (microneurography), cardiac baroreflex sensitivity (cross-spectral analysis and sequence method), sympathetic baroreflex sensitivity (threshold technique), plasma renin, and norepinephrine concentrations were measured. Measurements were performed under resting conditions, with and without electric baroreflex stimulation, for greater than or equal to 6 minutes during the same experiment. Intra-arterial BP was 193 +/- 9/94 +/- 5 mm Hg on medications. Acute electric baroreflex stimulation decreased SBP by 32 +/- 10 mm Hg (range of +7 to -108 mm Hg; p = 0.01). The depressor response was correlated with a muscle sympathetic nerve activity reduction (r(2) = 0.42; p < 0.05). In responders, muscle sympathetic nerve activity decreased sharply when ES started. Then, muscle sympathetic nerve activity increased but remained below the baseline level throughout the stimulation period. Heart rate decreased 4.5 +/- 1.5 beats/min with ES ($p < 0.05$). Plasma renin concentration decreased 20 +/- 8 % ($p < 0.05$). Electrical stimulation of carotid sinus baroreflex afferents acutely decreased arterial BP in hypertensive patients, without negative effects on physiological baroreflex regulation. The depressor response was mediated through sympathetic inhibition.

Sanchez et al (2010) evaluated carotid artery structural integrity after implantation of the CSLs. A total of 29 CSLs were implanted on the common carotid arteries of 8 sheep. The studies were terminated at 3 and 6 months post-implantation to assess anatomical and histological changes. Additionally, 10 patients with resistant hypertension were enrolled in the Rheos Multicenter Feasibility Trial. Duplex ultrasound (DUS) was performed before device implantation and at 1 and 4 months post-implantation in this patient cohort. An independent core laboratory assessed all DUSs. Ovine carotid angiography revealed no significant stenoses, while anatomical and histological evaluations demonstrated electrode encapsulation in a thin layer of connective tissue with no evidence of stenosis, erosion, or inflammation. Duplex ultrasound evaluation revealed no significant increase in peak systolic velocities of the common and internal carotid arteries 1 and 4 months after initial implantation, indicating a lack of injury, remodeling, or stenosis. The authors concluded that the current data suggested that the CSLs used with the Rheos System are not associated with the development of carotid stenosis or injury. These short-term data support the concept of CSL placement and merit long-term investigation in a larger multi-center prospective trial.

In a prospective, non-randomized, feasibility study, Scheffers et al (2010) examined the safety and effectiveness of Rheos therapy in drug-resistant hypertension patients. A total of 45 subjects with SBP greater than or equal to 160 mm Hg or DBP greater than or equal to 90 mm Hg despite at least 3 anti-hypertensive drugs were enrolled in this study. Subjects were followed-up for as long as 2 years. An external programmer was used to optimize and individualize efficacy. Baseline mean BP was 179/105 mm Hg and HR was 80 beats/min, with a median of 5 anti-hypertensive drugs. After 3 months of device therapy, mean BP was reduced by 21/12 mm Hg. This result was sustained in 17 subjects who completed 2 years of follow-up, with a mean reduction of 33/22 mm Hg. The device exhibited a favorable safety profile. The authors concluded that the Rheos device sustainably reduces BP in drug-resistant hypertensive subjects with multiple co-morbidities receiving numerous medications. They stated that this novel approach holds promise for patients with drug-resistant hypertension and is currently under evaluation in a prospective, placebo-controlled clinical trial.

Doumas et al (2009) stated that the role of the carotid baroreflex in BP regulation has been known for a long time but its effects were thought to be short-lived. Recent data indicate that stimulation of carotid baroreceptors may lower BP not only for short periods of time, but also in the long run. Recent advances in technology permitted the development of a new device (Rheos) that addresses problems with older devices. Several questions remain to be addressed before Rheos can be used widely, and several potential clinical applications remain to be clarified. The authors stated that the carotid baroreceptor reflex is probably not completely in control of BP. Baroreflexes are one of many control systems acting in concert. Joshi and associates (2009) stated that ES of the carotid sinus baroreceptor through a surgically implanted device is currently under clinical investigation and is showing some encouraging early results. Furthermore, Lovett et al (2009) noted that chronic results from feasibility studies indicated that Rheos therapy has an acceptable safety profile and may lead to long-term control of BP in drug-resistant hypertensive patients. Other effects include significant reductions in left ventricular mass and left atrial size. The spectrum of therapeutic impact suggests that Rheos therapy may improve long-term outcomes in drug-resistant hypertension and possibly benefit related populations. They noted that larger randomized, controlled trials are ongoing to verify chronic benefits.

Grassi et al (2010) stated that drug-resistant hypertension represents a condition frequently detected in clinical practice. Its main features are represented by its heterogeneous etiology as well as its very high cardiovascular risk. This latter peculiarity has implemented the research for new approaches to the treatment of the disease. These researchers discussed 2 of these approaches: (i) carotid baroreceptor ES and (ii) the renal denervation procedure. They stated that clinical studies and large-scale clinical trials are presently ongoing with the aim of defining the long-term safety and effectiveness of the 2 interventions. Taylor and Bisognano (2010) stated that chronic baroreceptor ES of the carotid sinus has been shown to reduce BP by inhibiting the sympathetic nervous system, especially the renal sympathetic tone. This finding has led to the development of implantable carotid sinus stimulators, which have now been studied in both animals and humans, as a means for treating chronic hypertension. The enthusiasm for this modality has led to ongoing studies, which will provide more information on its safety and effectiveness in patients with drug-resistant hypertension. The early study results using baroreflex ES therapy are promising and suggest that it may play a significant role in controlling BP in the future. Kougias et al (2010) stated that a model of ES of the carotid sinus has been developed and successfully tested in animals. Following these encouraging results, human trials to evaluate the clinical application of electrical carotid sinus manipulation in the treatment of systemic hypertension have commenced, and results so far indicated that this represents an exciting potential tool in the clinician's armament against chronic arterial hypertension.

Lohmeier and Iliescu (2011) summarized the pre-clinical studies that have provided insight into the mechanisms that account for the chronic BP-lowering effects of carotid baroreflex activation. Some of the mechanisms identified were predictable and confirmed by experimentation. Others have been surprising and controversial, and resolution will require further investigation. They stated that although feasibility studies have been promising, firm conclusions regarding the value of this device-based therapy for the treatment of drug-resistant hypertension awaits the results of current multi-center trials. Ng and colleagues (2011) noted that the Rheos Baroreflex Hypertension Therapy System is a new implantable device that can treat patients with hypertension resistant to multi-drug therapy, by activating the carotid baroreflex through ES of the carotid sinus wall. Recent studies in both normotensive and hypertensive canine models have reported sustained and clinically relevant reductions in arterial pressure and sympathetic activity with prolonged baroreflex activation. Clinical trials designed to evaluate the safety and effectiveness of this therapy in patients with drug-resistant hypertension, are now ongoing in both Europe and the United States.

Bisognano et al (2011) examined the effect of baroreflex activation therapy (BAT) on SBP in patients with resistant hypertension. This was a double-blind randomized trial of 265 subjects with resistant hypertension implanted and subsequently randomized (2:1) 1 month after implantation. Subjects received either BAT (group A) for the first 6 months or delayed BAT initiation following the 6-month visit (group B). The 5 co-primary endpoints were: (i) acute SBP responder rate at 6 months; (ii) sustained responder rate at 12 months; (iii) procedure safety; (iv) BAT safety; and (v) device safety. The trial showed significant benefit for the endpoints of sustained efficacy, BAT safety, and device safety. However, it did not meet the endpoints for acute responders or procedural safety. A protocol-specified ancillary analysis showed 42 % (group A) versus 24 % (group B) achieving SBP less than or equal to 140 mm Hg at 6 months (p = 0.005), with both groups achieving over 50 % at 12 months, at which point group B had received 6 months of BAT. The authors concluded that a clinically meaningful measure, those achieving a SBP of less than or equal to 140 mm Hg, yielded a significant difference between the groups. The weight of the overall evidence suggested that over the long-term, BAT can safely reduce SBP in patients with resistant hypertension. They stated that future clinical trials will address the limitations of this study and further define the therapeutic benefit of BAT.

In a single-arm, open-label study, Hoppe and associates (2012) evaluated the effectiveness of the Barostim neo™ system, a 2nd-generation system for delivering BAT. Subjects were patients with resistant hypertension (SBP greater than or equal to 140 mm Hg despite treatment with 3 or more medications, including 1 ore more diuretic). Stable medical therapy was required for 4 or more weeks before establishing pre-treatment baseline by averaging 2

SBP readings taken 24 or more hours apart. A total of 30 patients enrolled from 7 medical centers in Europe and Canada. From a baseline of 171.7 +/- 20.2/99.5 +/- 13.9 mm Hg, arterial BP decreased by 26.0 +/- 4.4/12.4 +/- 2.5 mm Hg at 6 months. In a subset (n = 6) of patients with prior renal nerve ablation, arterial BP decreased by 22.3 +/- 9.8 mm Hg. Background medical therapy for hypertension was unchanged during follow-up. Three minor procedure-related complications occurred within 30 days of implant. All complications resolved without sequelae. The authors concluded that BAT delivered with the second-generation system significantly lowers BP in resistant hypertension with stable, intensive background medical therapy, consistent with studies of the 1st-generation system for electrical activation of the baroreflex, and provides a safety profile comparable to a pace-maker. The findings of this small open-label study need to be validated by well-designed studies.

Jordan et al (2012) noted that recently, a novel implantable device was developed that produces an electrical field stimulation of the carotid sinus wall. Carefully conducted experiments in dogs provided important insight in mechanisms mediating the depressor response to electrical carotid sinus stimulation. Moreover, these studies showed that the treatment success may depend on the underlying pathophysiology of the hypertension. The authors stated that clinical studies suggested that electrical carotid sinus stimulation attenuates sympathetic activation of vasculature, heart, and kidney while augmenting cardiac vagal regulation, thus lowering BP; however, not all patients respond to treatment. They concluded that additional clinical trials are needed to ascertain the effectiveness of electrical carotid sinus stimulation in treatment resistant arterial hypertension.

Jordan et al (2013) stated that hypertension is the primary risk factor for cardiovascular and renal-disease endpoints. Medications help many patients but not all. Recently, 2 device-related treatments have been introduced, catheter-based renal denervation and electrical carotid sinus stimulation. Remuneration for these treatments is guaranteed in many countries even though basic information is missing. These investigators drew attention to deficiencies in the database. For catheter-based renal denervation, few large-animal data are available to investigate the effect of the intervention on the histology of the arterial wall. No functional data are available regarding re-innervation. For carotid sinus stimulation, the situation is similar. The authors concluded that acute activation of either treatment seems to reduce sympathetic tone dramatically; however, whether or not the effects are sustained over time is unknown. They stated that no "end-point" data are available for either treatment, and devices should be subjected to evidence-based standards before widespread introduction.

Chobanyan-Jurgens et al (2015) stated that treatment-resistant arterial hypertension is associated with excess cardiovascular morbidity and mortality. Electrical carotid sinus stimulators engaging baroreflex afferent activity have been developed for such patients. Indeed, baroreflex mechanisms contribute to long-term BP control by governing efferent sympathetic and parasympathetic activity. The first-generation carotid sinus stimulator applying bilateral bipolar stimulation reduced BP in a controlled clinical trial but nevertheless failed to meet the primary efficacy end-point. The second-generation device utilizes smaller unilateral unipolar electrodes, thus decreasing invasiveness of the implantation while saving battery. An uncontrolled clinical study suggested improvement in BP with the 2nd-generation device. The authors hoped that these findings as well as preliminary observations suggesting cardiovascular and renal organ protection with electrical carotid sinus stimulation will be confirmed in properly controlled clinical trials. Moreover, these researchers noted that it is important to find ways to better identify patients who are most likely to benefit from electrical carotid sinus stimulation.

Victor (2015) noted that arterial baroreceptors are mechano-sensitive sensory nerve endings in the walls of the carotid sinuses and aortic arch that buffer the increases and decreases in arterial BP. Electrical field stimulation of the carotid sinus, known as carotid baroreflex activation therapy, holds promise as a novel device-based intervention to supplement, but not replace, drug therapy for patients with resistant hypertension. Acute electrical field stimulation of even 1 carotid sinus can cause a sufficiently large reflex decrease in BP to overcome off-setting reflexes from the contralateral carotid baroreceptors and aortic baroreceptors that are not paced. However, the initial phase III Rheos Pivotal Trial on continuous carotid baroreceptor pacing for resistant hypertension with the first-generation baroreceptor pacemaker yielded equivocal data on efficacy and adverse effects due to facial nerve injury during surgical implantation. A miniaturized second-generation pacing electrode has seemingly overcome the safety issue, and early results with the new device suggested efficacy of unilateral carotid sinus stimulation in heart failure (HF). The authors stated that a phase III clinical trial of this new device for resistant hypertension has been registered.

Heusser and colleagues (2016) stated that bilateral bipolar electric carotid sinus stimulation acutely reduced muscle sympathetic nerve activity (MSNA) and BP in patients with resistant arterial hypertension; but is no longer available. The 2nd-generation device uses a smaller unilateral unipolar disk electrode to reduce invasiveness while saving battery life. These researchers hypothesized that the 2nd-generation device acutely lowers BP and MSNA in treatment-resistant hypertensive patients. In this study, a total of 18 treatment-resistant hypertensive patients (9 women/9 men; 53 ± 11 years; 33 ± 5 kg/m(2)) on stable medications have been included in the study. These investigators monitored finger and brachial BP, HR, and MSNA. Without stimulation, BP was 165 ± 31/91 ± 18 mm Hg, HR was 75 ± 17 bpm, and MSNA was 48 ± 14 bursts/min. Acute stimulation with intensities producing side effects that were tolerable in the short-term elicited inter-individually variable changes in systolic BP (-16.9 ± 15.0 mm Hg; range of 0.0 to -40.8 mm Hg; $p = 0.002$), HR (-3.6 ± 3.6 bpm; $p = 0.004$), and MSNA (-2.0 ± 5.8 bursts/min; $p = 0.375$). Stimulation intensities had to be lowered in 12 patients to avoid side effects at the expense of efficacy (systolic BP, -6.3 ± 7.0 mm Hg; range of 2.8 to -14.5 mm Hg; $p = 0.028$ and HR, -1.5 ± 2.3 bpm; $p = 0.078$; comparison against responses with side effects). Reductions in diastolic BP and MSNA (total activity) were correlated (r(2) = 0.329; $p = 0.025$). The authors concluded that in this patient cohort, unilateral unipolar electric baroreflex stimulation acutely lowered BP. However, side effects may limit efficacy. They stated that this approach should be tested in a controlled comparative study.

Wallbach and associates (2016) noted that BAT has been demonstrated to decrease office BP in the randomized, double-blind Rheos Trial. There are limited data on 24-hour BP changes measured by ambulatory BP measurements (ABPMs) using the 1st generation Rheos BAT system suggesting a significant reduction but there are no information about the effect of the currently used, unilateral BAT neo device on ABPM. In an observational study, ABPM was performed before BAT implantation and 6 months after initiation of BAT in patients treated with the BAT neo device for uncontrolled resistant hypertension. A total of 51 patients were included in this study, 7 dropped-out from analysis because of missing or insufficient follow-up. After 6 months, 24-hour ambulatory systolic (from 148 ± 17 mm Hg to 140 ± 23 mm Hg, $p < 0.01$), diastolic (from 82 ± 13 mm Hg to 77 ± 15 mm Hg, $p < 0.01$), day- and night-time systolic and diastolic BP (all $p \leq 0.01$) significantly decreased while the number of prescribed anti-hypertensive classes could be reduced from 6.5 ± 1.5 to 6.0 ± 1.8 ($p = 0.03$); HR and pulse pressure remained unchanged. Baroreflex activation therapy was equally effective in reducing ambulatory BP in all subgroups of patients. The authors concluded that this was the first study demonstrating a significant BP reduction in ABPM in patients undergoing chronically stimulation of the carotid sinus using the BAT neo device. About that BAT-reduced office BP and improved relevant aspects of ABPM, BAT might be considered as a new therapeutic option to reduce cardiovascular risk in patients with resistant hypertension. Moreover, they stated that randomized controlled trials (RCTs) are needed to evaluate BAT effects on ABPM in patients with resistant hypertension accurately.

Ng and colleagues (2016) discussed 7 novel devices for the treatment of resistant hypertension (RHTN): (i) renal nerve denervation, (ii) BAT (carotid sinus stimulation), (iii) carotid body ablation, (iv) central iliac arterio-venous anastomosis, (v) deep brain stimulation, (vi) median nerve stimulation, and (vii) vagal nerve stimulation. The authors noted that the Barostim Hypertension Pivotal Trial (clinicaltrials.gov: NCT01679132) is currently in progress and aims to enroll 310 patients with RHTN randomized to receiving optimal medical management alone or in combination with the BAT, which may also have a role outside of BP management and is currently being evaluated as an adjunctive therapy in HF.

Jordan (2017) stated that device-based anti-hypertensive treatments have primarily been developed and clinically tested for patients with hypertension refractory to pharmacological treatment. Most but not all device-based treatments target the sympathetic nervous system and provided important new insight in the mechanisms of human hypertension. This investigator provided an overview on the scientific rational and clinical data on recent device-based anti-hypertensive treatment approaches. Device-based treatments targeting the sympathetic nervous system include catheter-based renal nerve ablation, electrical carotid sinus stimulation, modulation of baroreflex transduction through a dedicated carotid stent, carotid body denervation, and deep brain stimulation. Creation of a defined arterio-venous stent with a coupler device and removal of stimulatory antibodies against alpha adrenoreceptors have also been tested. The author concluded that the clinical evidence differed from therapy to therapy with the largest data-set for renal nerve ablation followed by electrical carotid sinus stimulation; however, none has been proven effective in sham-controlled clinical trials, and none has been shown to reduce cardiovascular morbidity or mortality. The author stated that before effectiveness is proven, these treatments should not be part of routine medical care and only be applied in the setting of clinical studies.

An assessment by the Ludwig Boltzmann Institute for Health Technology Assessment of baroreceptor activation therapy for treatment-resistant hypertension (Hawlick and Winkler, 2018) concluded that the quality of evidence was "very low."

Gierthmuehlen and colleagues (2020) presented an overview on recent developments in permanent implant-based therapy of resistant hypertension. The American Heart Association (AHA) recently updated their guidelines to treat hypertension. As elevated BP now is defined as a systolic BP of above 120 mmHg, the prevalence of hypertension in the U.S. has increased from 32 % (old definition of hypertension) to 46 %. In the past years, device- and implant-mediated therapies have evolved and extensively studied in various patient populations. Despite an initial drawback in a RCT of bilateral carotid sinus stimulation (CSS), new and less invasive and unilateral systems for BAT with the BAROSTIM NEO have been developed which show promising results in small non-RCTs. Selective vagal nerve stimulation (VNS) has been successfully evaluated in rodents, but has not yet been tested in humans. A new endovascular approach to re-shape the carotid sinus to lower BP (MobiusHD) has been introduced (baroreflex amplification therapy) with favorable results in non-RCT trials. However, long-term results are not yet available for this therapeutic option. A specific subgroup of patients, those with indication for a 2-chamber cardiac pacemaker, may benefit from a new stimulation paradigm which reduces the atrioventricular (AV) latency and thus limits the filling time of the left ventricle. The most invasive approach for resistant hypertension still is the neuromodulation by deep brain stimulation (DBS), which has been shown to significantly lower BP in single cases. The authors concluded that implant-mediated therapy remains a promising approach for the treatment of resistant hypertension. Due to their invasiveness, such therapeutic options must prove superiority over conventional therapies with regard to safety and efficacy before they could be generally offered to a wider patient population. Overall, BAROSTIM NEO and MobiusHD, for which large RCTs will soon be available, are likely to meet those criteria and may represent the first implant-mediated therapeutic options for hypertension, while the use of DBS probably will be reserved for individual cases. The utility of VNS awaits appropriate assessment.

Wallbach et al (2023) stated that a relevant number of patients with resistant hypertension do not achieve blood pressure (BP) dipping during night-time. This inadequate nocturnal BP reduction is associated with elevated cardiovascular risks. In a prospective, observational study, these researchers examined if a night-time intensification of BAT might improve nocturnal BP dipping. Non-dippers treated with BAT for at least 6 months were included. BAT programming was modified in a 2-step intensification of night-time stimulation at baseline and week 6. Twenty-four hours ABPMs were conducted at inclusion and after 3 months. A total of 24 patients with non- or inverted-dipping pattern, treated with BAT for a median of 44 months

(IQR 25 to 52) were included. At baseline of the study, patients were 66 ± 9 years old, had a BMI of 33 ± 6 kg/m2 , showed an office BP of 135 ± 22/72 ± 10 mmHg, and took a median number of anti-hypertensives of 6 (inter-quartile range [IQR] 4 to 9). Night-time stimulation of BAT was adapted by an intensification of pulse width from 237 ± 161 to 267 ± 170 μs ($p = 0.003$) while frequency ($p = 0.10$) and amplitude ($p = 0.95$) remained unchanged. Up-titration of BAT programming resulted in an increase of systolic dipping from 2 ± 6 % to 6 ± 8 % ($p = 0.03$) accompanied with a significant improvement of dipping pattern ($p = 0.02$); and 24-hour ABPM, day- and night-time ABP remained unchanged. The authors concluded that programming of an intensified night-time BAT interval improved dipping profile in patients treated with BAT, while the overall 24-hour ABP did not change. These researchers stated that whether the improved dipping response would contribute to a reduction of cardiovascular risk beyond the BP-lowering effects of BAT, however, remains to be determined.

The authors stated that a main drawback of this study was the small number of patients (n = 24), which was the consequence of the current availability and diffusion of BAT in the treatment of resistant hypertension; thus, these findings should be interpreted with caution, especially with regard to the lack of significant changes. In fact, these results might primarily serve for hypotheses-generation. Larger studies are needed to confirm these data. The study also lacked randomization, a control group, and blinding. However, the apparently greater reduction in night-time BP in non-dippers may be, at least in part, due to the effect of regression to the mean. A prospective, randomized, sham-controlled study examining BAT in patients with resistant hypertension using 24-hour ABP with nocturnal dipping status as secondary endpoint could overcome the afore-mentioned drawbacks. Another relevant source for bias was the adherence to anti-hypertensive drugs. This trial did not examine the impact of adherence to anti-hypertensive medication. Changes in adherence might have influenced BP and dipping response throughout the study. Another drawback was that no measures of indices of sympathetic activity, such as urinary catecholamines was carried out in this trial.

Hering and Narkiewicz (2024) noted that uncontrolled hypertension drives the global burden of increased cardiovascular disease (CVD) morbidity and mortality. Although high BP is treatable and preventable, only 50 % of the patients with hypertension undergoing treatment have their BP controlled. The failure of poly-pharmacy to attain adequate BP control may be due to a lack of physiological response; however, medication non-adherence and clinician inertia to increase treatment intensity are critical factors associated with poor hypertension management. The long-time medication titration, lifelong drug therapy, and often multi-drug treatment strategy are frustrating when the BP goal is not attained, resulting in increased CVD risk and a substantial burden on the healthcare system. Growing evidence indicates that neurohumoral activation is critical in initiating and maintaining elevated BP and its adverse consequences. Over the past decades, device-based therapies targeting the mechanisms underlying hypertension pathophysiology have been extensively studied. Among these, robust clinical experience for hypertension management exists for renal denervation (RDN), BAT, carotid body denervation (CBD), central arterio-venous anastomosis, and to a lesser extent, DBS. The authors concluded that further investigations are needed to define the role of device-based approaches as an alternative or adjunctive therapeutic option for the treatment of refractory hypertension.

Biffi et al (2024) stated that in resistant hypertensive patients acute carotid baroreflex stimulation is associated with a BP reduction, believed to be mediated by a central sympatho-inhbition; however, the evidence for this sympatho-modulatory effect is limited. In a systematic review and meta-analysis, these investigators examined the sympatho-modulatory effects of acute carotid baroreflex stimulation in drug-resistant and uncontrolled hypertension, based on the results of microneurographic studies. The analysis included 3 studies examining MSNA and examining 41 resistant uncontrolled hypertensives. The evaluation included assessment of the relationships between MSNA and clinic HR and BP changes associated with the procedure. Carotid baroreflex stimulation induced an acute reduction in clinic SBP and diastolic BP (DBP), which achieved statistical significance for the former variable only (SBP: -19.98 mmHg, 90 % CI: -30.52 to -9.43, $p < 0.002$), (DBP: -5.49 mmHg, 90 % CI: -11.38 to

0.39, p = NS). These BP changes were accompanied by a significant MSNA reduction (-4.28 bursts/min, 90 % CI: -8.62 to 0.06, $p < 0.07$), and by a significant HR decrease (-3.65 beats/min, 90 % CI: -5.49 to -1.81, $p < 0.001$]. No significant relationship was detected between the MSNA, SBP, and DBP changes induced by the procedure, this being the case also for HR. The authors concluded that these findings showed that the acute BP-lowering responses to carotid baroreflex stimulation, although associated with a significant MSNA reduction, were not quantitatively related to the sympatho-moderating effects of the procedure; suggesting that these BP effects depended only in part on central sympatho-inhibition, at least in the acute phase following the intervention.

Magnetic Stimulation of Carotid Sinus

Zhang and colleagues (2017) evaluated the effectiveness of magnetic stimulation of carotid sinus (MSCS), a non-invasive strategy, for lowering BP in rabbits; MSCS with graded intensities and frequencies were systematically attempted in normotensive rabbits; BP was recorded dynamically. Sino-aortic denervation and plasma hormone level analyses were performed. When the right carotid sinus was stimulated at 1-Hz frequency, a dose-effect relationship was observed between stimulation intensity (100 to 250 % motor threshold) and mean arterial pressure (MAP) decrement (3.6±1.0 to 10.4±2.3mmHg). When stimulation intensity was fixed at 200 % motor threshold, the median reduction of MAP in 1-Hz group [10.8 (8.6 to 14.9)mmHg] was significantly higher than that in other frequency groups (all $p<0.05$). Heart rates declined transiently after the initiation of MSCS. Compared with baseline, plasma epinephrine level increased during MSCS (from 33.9±5.5pg/ml to 88.1±9.6, $p=0.002$). After ipsilateral sino-aortic denervation, MAP decrement was remarkably blunted compared with that in sham animals (7.0±0.8mmHg versus 13.0±1.1mmHg, $p=0.001$). The authors concluded that the findings of the current study demonstrated that MSCS treatment can lower the arterial pressure in normotensive rabbits. They stated that is preliminarily finding warrants further studies to establish the effectiveness of MSCS in treating refractory hypertension.

Jordan et al (2024) stated that an imbalance between cardiovascular para-sympathetic and sympathetic activity towards sympathetic predominance has been implicated in the pathogenesis of treatment-resistant arterial hypertension and HF. Arterial baroreceptors control efferent cardiovascular autonomic activity; thus, they have been recognized as potential treatment targets. Baroreflex activation therapy via electrical carotid sinus stimulation is a device-based approach to modulate cardiovascular autonomic activity. Electrical carotid sinus stimulation lowered BP in various hypertensive animal models and improved cardiac re-modeling and survival in pre-clinical models of HF. In human mechanistic profiling studies, electrical carotid sinus stimulation lowered BP via sympathetic inhibition with substantial inter-individual variability. The 1st-generation device reduced BP in controlled and uncontrolled clinical trials. Controlled clinical trials proving effectiveness in BP reduction in patients with hypertension do not exist for the currently available 2nd-generation carotid sinus stimulator. Investigations in HF patients showed improved symptoms, QOL, as well as natriuretic peptide biomarkers. The authors concluded that electrical carotid sinus stimulation is an interesting technology to modulate cardiovascular autonomic control; however, the contribution of sympathetic activity to BP in hypertensive patients is highly variable. Therefore, patients showed large variability in the BP-lowering effect of treatments attenuating sympathetic activity. Thus, it is expected that patients on BAT via electrical carotid sinus stimulation also exhibit a variable response. It is still impossible, to prospectively define patients who are more or less likely to respond. In fact, it is not even possible to differentiate poor responses secondary to sub-optimal electrode placement from true non-responders. The state-of-affairs may not be acceptable considering the costs and invasiveness of the intervention. Moreover, proper dose titration may be especially important in patients with HF. Excess reduction in sympathetic activity may increase mortality in HF patients as evidenced by the MOXCON Trial (Cohn et al, 2003). Lastly, data from controlled trials with hard clinical end-points are needed to justify more widespread use.

Other Indications

Heart Failure

Georgakopoulos et al (2011) stated that heart failure with preserved ejection fraction (HFpEF) is a substantial public health issue, equal in magnitude to heart failure with reduced ejection fraction. Clinical outcomes of HFpEF patients are generally poor, and no therapy has been shown to be effective in randomized clinical trials. Baroreflex activation therapy (BAT) produced by stimulating the carotid sinuses using the Rheos device is being studied for the treatment of hypertension, the primary co-morbidity of HFpEF. Other potential benefits include regression of left ventricular hypertrophy, normalization of the sympatho-vagal balance, inhibition of the renin-angiotensin-aldosterone system, arterio- and veno-dilation, and preservation of renal function. The authors concluded that BAT may be a promising therapy for HFpEF and introduced the HOPE4HF trial, a randomized outcomes trial designed to evaluate the clinical safety and effectiveness of BAT in the HFpEF population.

The Institute for Clinical Systems Improvement's clinical practice guideline on "Heart failure in adults" (ICSI, 2011) did not mention the use of BAT.

An European Heart Rhythm Association’s report on "New devices in heart failure" (Kuck et al, 2014) stated that several new devices for the treatment of HF patients have been introduced and are increasingly used in clinical practice or are under clinical evaluation in either observational and/or randomized clinical trials. These devices include cardiac contractility modulation, spinal cord stimulation, carotid sinus nerve stimulation, cervical vagal stimulation, intra-cardiac atrioventricular nodal vagal stimulation, and implantable hemodynamic monitoring devices.

Abraham et al (2015) stated that increased sympathetic and decreased parasympathetic activity contribute to heart failure (HF) symptoms and disease progression. Baroreflex activation therapy (BAT) results in centrally mediated reduction of sympathetic outflow and increased parasympathetic activity. In a clinical trial, these researchers examined the safety and effectiveness of carotid BAT in advanced HF. Patients with New York Heart Association (NYHA) functional class III HF and ejection fractions (EFs) of 35 % or less on chronic stable guideline-directed medical therapy (GDMT) were enrolled at 45 centers in the U.S., Canada, and Europe. They were randomly assigned to receive ongoing GDMT alone (control group) or ongoing GDMT plus BAT (treatment group) for 6 months. The primary safety endpoint was system- and procedure-related major adverse neurological and cardiovascular events. The primary efficacy endpoints were changes in NYHA functional class, quality-of-life (QOL) score, and 6-minute hall walk (6MHW) distance. A total of 146 patients were randomized, 70 to control and 76 to treatment. The major adverse neurological and cardiovascular event-free rate was 97.2 % (lower 95 % confidence bound 91.4 %). Patients assigned to BAT, compared with control group patients, experienced improvements in the 6MHW distance (59.6 ± 14 m versus 1.5 ± 13.2 m; $p = 0.004$), QOL score (-17.4 ± 2.8 points versus 2.1 ± 3.1 points; $p < 0.001$), and NYHA functional class ranking ($p = 0.002$ for change in distribution). BAT significantly reduced N-terminal pro-brain natriuretic peptide ($p = 0.02$) and was associated with a trend toward fewer days hospitalized for HF ($p = 0.08$). The authors concluded that BAT was safe and improved functional status, QOL, exercise capacity, N-terminal pro-brain natriuretic peptide, and possibly the burden of HF hospitalizations in patients with GDMT-treated NYHA functional class III HF. Moreover, these researchers stated that this latter observation should be confirmed in an adequately powered prospective outcome trial.

The authors stated that the relatively small number of patients studied may limit the interpretation of some of the results of this study. Another potential limitation may be the lack of patient blinding and a sham control, leading to a “placebo effect” in the treatment arm, or a lack of blinding in investigator assessment of endpoints, leading to bias. However, the magnitude of improvement in the primary endpoints was substantially larger than that attributable to such a placebo effect or bias in prior device trials. For example, in prior studies of cardiac resynchronization therapy (CRT), the implantation of an inactive device was associated with a 10-m improvement in 6MHW distance, a placebo effect that fell far short of the nearly 60-m improvement observed with BAT in the present study. Furthermore, at least 1 of the endpoints significantly improved by BAT, NT-proBNP, was not prone to a placebo effect. The difference in follow-up schedule between study groups outside the United States (OUS) had the potential to bias the results. However, there were no statistically significant differences in the treatment effect between the U.S. and OUS subjects. Similarly, the differential collection of hospitalization data by world region could also introduce bias.

Weaver et al (2016) described the intra-operative experience along with long-term safety and effectiveness of the 2nd-generation BAT system in patients with HF and reduced EF HF (HFrEF). In a randomized trial of NYHA Class III HFrEF, 140 patients were assigned 1:1 to receive BAT plus medical therapy or medical therapy alone. Procedural information along with safety and effectiveness data were collected and analyzed over 12 months. Within the cohort of 71 patients randomized to BAT, implant procedure time decreased with experience, from 106 ± 37 mins on the 1st case to 83 ± 32 mins on the 3rd case. The rate of freedom from system- and procedure-related complications was 86 % through 12 months, with the percentage of days alive without a complication related to system, procedure, or underlying cardiovascular condition identical to the control group. The complications that did occur were generally mild and short-lived. Overall, 12 months therapeutic benefit from BAT was consistent with previously reported effectiveness through 6 months: there was a significant and sustained beneficial treatment effect on NYHA functional Class, QOL, 6MHW distance, plasma N-terminal pro-brain natriuretic peptide (NT-proBNP), and systolic blood pressure. This was true for the full trial cohort and a pre-defined subset not receiving cardiac resynchronization therapy (CRT). There was a rapid learning curve for the specialized procedures entailed in a BAT system implant. The authors concluded that the BAT system implantation was safe with the therapeutic benefits of BAT in patients with HFrEF being substantial and maintained for at least 1 year. These researchers stated that the level of evidence supporting BAT in HFrEF and the maturity of the 2nd-generation system indicated that the necessary conditions are in place for an outcomes trial. They noted that a pivotal outcomes trial of BAT in HFrEF is expected to commence in 2016.

These investigators stated that although the phase-I and phase-II clinical trials both confirmed the long-term benefits of BAT in HFrEF, the present knowledge base needs to be expanded. The phase-I trial was open-label and consisted of 11 patients whereas the phase-II trial consisted of 140 patients and used a medical management control group. Therefore, the magnitude of benefit relative to sham was undetermined. Reductions in muscle sympathetic nerve activity (phase-I) and NT-proBNP (phase-II) provided objective evidence that corroborated therapy benefit even in the absence of control. Although these findings bode well for an outcomes trial, actual results should be accrued on a significant number of patients to confirm reductions in cardiovascular mortality and HF hospitalization. This is the objective of a pivotal trial planned to begin enrollment in 2016. Although the BAT system and implant techniques have evolved in a significantly positive direction from the 1t generation, opportunities for further improvement exist. Because the most common electrode locations are known and 1 region in particular has become a de facto starting position for mapping, it is conceivable that an electrode could be developed to cover the locations where a response is most likely. If anatomical landmarks coupled with a versatile electrode design were to obviate the need for mapping, not only would procedure time diminish, but also the complexity of anesthesia would be substantially reduced. Furthermore, despite the fact that the suturing technique required to fix the BAT electrode is well within the capabilities of a vascular or cardiac surgeon, the process can be time-consuming. If the number of required sutures could be reduced through a design change or the use of adjunctive fixatives or both, the procedure would also be simplified and shortened. BAT is presently available for commercial use in resistant hypertension and HFrEF in regions that accept evidence commensurate with CE Marking. Investigational plans are also in place to extend BAT availability for those conditions in the U.S. Meanwhile, evidence suggested that BAT may be a useful treatment in chronic kidney disease, and HF with preserved EF.

In a systematic review, Schmidt et al (2020) examined the available data in the literature regarding the safety and effectiveness of BAT in the treatment of heart failure with reduced ejection fraction (HFrEF) patients. These investigators searched electronic databases including PubMed, Embase, CENTRAL, Scopus, and Web of Science using Mesh and free terms for heart failure and BAT. No language restriction was employed for the searches. They included full peer-reviewed publications of clinical studies (randomized or not), including patients with HFrEF undergoing BAT, with or without control group, assessing safety and effectiveness outcomes. One reviewer conducted the analysis of the selected abstracts and the full-text articles, carried out data extraction, and evaluated the methodological quality of the selected articles. The methodological quality was examined according to the Cochrane Collaboration instruments. A descriptive summary of the results was provided. Of the 441 citations screened, 10 publications were included (3 were only conference abstracts), reporting data from 3 studies. Only 1 study was a randomized clinical trial; 2 studies reported a 6-month follow-up, and the other study analyzed outcomes up to 41 months. The procedure appeared to be safe when carried out by a well-trained, multi-professional team. An 86 % rate of system and procedure-related complication-free was reported, with no cranial nerve injuries. Improvements were observed in NYHA class of HF, quality of life (QOL), 6 min walk test (6MWT), and hospitalization rates, as well as in muscle sympathetic nerve activity. No meta-analysis was performed because of the lack of homogeneity across studies; the results from each study were reported individually. The authors concluded that BAT procedure appeared to be safe if appropriate training is provided. Improvements in clinical outcomes were described in all included studies; however, several limitations did not allow these investigators to make conclusive statements on the effectiveness of BAT for HFrEF. They stated that further high-quality RCTs with long-term follow-up are still needed to obtain conclusive results.

The authors stated that this study had several drawbacks. Only 3 studies were available in the literature answering the review research question, and only 1 of them was designed as an RCT. Furthermore, 3 of the 10 included publications were only abstracts published in conferences and not peer-reviewed publications. The small sample size and the heterogeneity across the included studies lacked the power to make conclusive statements. More importantly, these researchers designed their study as a rapid systematic review. They performed all the steps of a traditional systematic review; however, they were executed by only 1 reviewer. This approach has been previously described in the literature, and its value has been recognized by important organizations, such as the Cochrane Collaboration. A rapid review has a shorten time of execution, making it possible to obtain results timely and with reduced costs.

Zile and colleagues (2020) examined the safety and effectiveness of BAT in patients with HF with reduced ejection fraction (HFrEF). The BeAT-HF (Baroreflex Activation Therapy for Heart Failure) trial was a prospective, multi-center RCT; subjects were randomized 1:1 to receive either BAT plus optimal medical management (BAT group) or optimal medical management alone (control group). A total of 4 patient cohorts were created from 408 randomized patients with HFrEF using the following enrollment criteria: current New York Heart Association (NYHA) functional class III or functional class II (patients who had a recent history of NYHA functional class III); EF of less than or equal to 35 %; stable medical management for greater than or equal to 4 weeks; and no Class I indication for cardiac re-synchronization therapy. Effectiveness end-points were the change from baseline to 6 months in 6-min hall walk distance (6MHW), Minnesota Living with HF Questionnaire quality-of-life (QOL) score, and N-terminal pro-B-type natriuretic peptide (NT-proBNP) levels. The safety end-point included the major adverse neurological or cardiovascular system or procedure-related event rate (MANCE). Results from, timeline and rationale for, cohorts A, B, and C were presented in detail in the text. Cohort D, which represented the intended use population that reflected the Food and Drug Administration (FDA)-approved instructions for use (enrollment criteria plus NT-proBNP of less than 1,600 pg/ml), consisted of 245 patients followed-up for 6 months (120 in the BAT group and 125 in the control group). BAT was safe and significantly improved QOL, 6MHW, and NT-proBNP. In the BAT group versus the control group, QOL score decreased ($\Delta = -14.1$; 95 % confidence interval [CI]: -19 to -9; $p < 0.001$), 6MHW distance increased ($\Delta = 60$ m; 95 % CI: 40 to 80 m; $p < 0.001$), NT-proBNP decreased ($\Delta = -25$ %; 95 % CI: -38 % to -9 %; $p = 0.004$), and the MANCE-free rate was 97 % (95 % CI: 93 % to 100 %; $p < 0.001$). The authors concluded that BAT was safe and significantly improved QOL, exercise capacity, and NT-proBNP. Moreover, these researcher stated that further studies are needed to examine the impact of BAT on the frequency of hospitalization and mortality, and identify patients with HFrEF most likely to gain lasting benefit from this type of intervention.

The authors stated that this study had several drawbacks. The BeAT-HF trial pre-market phase did not examine morbidity and mortality or change in cardiovascular structure or function end-points. Data from previous studies suggested that the BAT-induced reduction in the NT-proBNP data of 25 % made it highly probable that morbidity and mortality would also be reduced, and that structural and functional re-modeling would occur with BAT. For example, in the PARADIGM-HF (Efficacy and Safety of LCZ696 Compared to Enalapril on Morbidity and Mortality of Patients With Chronic Heart Failure) trial, morbidity and mortality were reduced when NT-proBNP fell by as little as 10 %, regardless of the treatment groups (sacubitril/valsartan versus enalapril). The GUIDE-IT (Guiding Evidence Based Therapy Using Biomarker Intensified Treatment) trial suggested that reduction in NT-proBNP from greater than 1,000 to less than 1,000 pg/ml was associated with a significant improvement in LV systolic function (increased EF) and LV re-modeling (reduced LV end-diastolic volume). However, all these specific end-points will require additional studies. Heart failure hospitalization and cardiovascular mortality rates will be examined in the post-market phase of BeAT-HF. Enrollment will continue as initially planned until a total of 480 patients have been randomized. The post-market phase is intended to expand the indication of use to reduction of HF hospitalizations and cardiovascular mortality. This post-market phase will be achieved when 320 mortal and morbid events have occurred. A supplemental pre-market approval will then be submitted to the FDA. Moreover, these researchers noted that BeAT-HF was not a blinded trial; the control group did not have an implanted BAT device. It was clearly acknowledged that 6MHW, QOL, NYHA functional class might be subject to placebo effects. This was why the NT-proBNP data served a pivotal role in supporting the results of these patient-centered symptomatic end-points.

Furthermore, an UpToDate review on "Overview of the management of heart failure with reduced ejection fraction in adults" (Colucci, 2020) does not mention implantable carotid sinus stimulator as a management / therapeutic option.

Malangu et al (2021) noted that there has been significant interest in research for the development of device-based therapy as a therapeutic option of HF, whether it is with reduced or preserved EF. This is due to the high morbidity and mortality rate in patients with HF despite recent advances in pharmacotherapies. Following the success of cardiac re-synchronization therapy, baroreceptor activation therapy (BAT) has emerged as another novel device-based treatment for HF. The Barostim neo was developed by CVRx (Minneapolis, MN) for the treatment of mild-to-severe HF. The device works by electrically activating the baroreceptor reflex with the goal to restore the maladaptive autonomic imbalance that is observed in patients with HF. The authors concluded that preliminary clinical investigations had given promising results with an encouraging safety profile. Moreover, these researchers stated that BAT as a therapeutic option for HF is still investigational at this time.

The American College of Cardiology/American Heart Association/Heart Failure Society of America (ACC/AHA/HFSA)'s guideline for the management of heart failure (Heidenreich et al, 2022) stated that "Trials of device stimulation of the vagus nerve, spinal cord, and baroreceptors have had mixed responses. An implantable device that electrically stimulates the baroreceptors of the carotid artery has been approved by the FDA for the improvement of symptoms in patients with advanced HF who are unsuited for treatment with other HF devices including CRT. In a prospective, multicenter, RCT with a total of 408 patients with current or recent NYHA class III HF, LVEF ≤ 35%, baroreceptor stimulation was associated with improvements in QOL, exercise capacity, and NT-proBNP levels. To date, there are no mortality or hospitalization rates results available with this device".

Lindenfeld et al (2021) examined sex differences in the safety and effectiveness of BAT in the BeAT-HF (Baroreflex Activation Therapy for Heart Failure) Trial. Patients were randomized 1:1 to receive GDMT alone (control group) or BAT plus GDMT. Pre-specified subgroup analyses including change from baseline to 6 months in 6MWD, QOL assessed using the Minnesota Living with Heart Failure Questionnaire (MLWHQ), NYHA functional class, and NT-proBNP were conducted in men versus women. A total of 53 women and 211 men were evaluated. Women had similar baseline NT-proBNP levels, 6MWDs, and percentage of subjects with NYHA functional class III symptoms but poorer MLWHQ scores (mean of 62 ± 22 versus 50 ± 24; $p = 0.01$) compared with men. Women experienced significant improvement from baseline to 6 months with BAT plus GDMT relative to GDMT alone in MLWHQ score (-34 ± 27 versus -9 ± 23, respectively; $p < 0.01$), 6MWD (44 ± 45 m versus -32 ± 118 m; $p < 0.01$), and improvement in NYHA functional class (70 % versus 27 %; $p < 0.01$), similar to the responses observed in men, with no significant difference in safety. Women receiving BAT plus GDMT had a significant decrease in NT-proBNP (-43 % versus 7 % with GDMT alone; difference -48 %; $p < 0.01$), while in men this decrease was -15 % versus 2 %, respectively (difference -17 %; $p = 0.08$), with an interaction p value of 0.05. The authors concluded that women in BeAT-HF had poorer baseline QOL than men but demonstrated similar improvements with BAT in 6MWD, QOL, and NYHA functional class. Women had a significant improvement in NT-proBNP, whereas men did not. These researchers stated that these preliminary findings were consistent with the response observed by sex to other GDMTs as well as CRT and suggested that women are likely to benefit from BAT at least as much as men.

The authors stated that the main drawback of this study was the small number of women (n = 53) enrolled in BeAT-HF relative to men, as these comparisons were not prospectively powered. In many symptomatic domains (QOL score, exercise capacity, and functional status), women improved with BAT more than men; however, the difference was not statistically significant, and it was possible that these differences in response might become significant in larger trials. In fact, the interaction p value between men and women became significant for QOL and NT-proBNP when the phase-2 data were included. Women receiving BAT plus GDMT, however, had significant decreases in NT-proBNP between treatment groups from baseline to 6 months, whereas in men, the decrease was not significant, except when phase-2 data were included. Although it is certainly possible that this difference may be spurious, NT-proBNP is a robust marker in this study, as it is both objective (in an unblinded trial) and a generally strong predictor of HF outcomes, as described. These researchers stated that larger studies are needed to confirm this finding. Finally, the BeAT-HF Trial thus far has not examined endpoints in morbidity and mortality or HF hospitalization; these are being studied in the now completed trial. Because baseline sex differences were established in a host of patient-centered outcomes, this prompts questions regarding whether women who are more symptomatic may be more likely to be identified and enrolled in clinical trials with a novel therapy such as BAT. Furthermore, these baseline differences appeared across multiple facets of distress, from HF symptoms to anxiety and depressive reports. Fortunately, these baseline differences were not sustained across the course of the trial, and women experienced comparable benefits from the therapy.

Coats et al (2022) noted that heart failure with reduced ejection fraction (HFrEF) remains associated with high morbidity and mortality, poor QOL and significant exercise limitation. Sympatho-vagal imbalance has been shown to predict adverse prognosis and symptoms in HFrEF, yet it has not been specifically targeted by any guideline-recommended device therapy to-date. Barostim, which directly addresses this imbalance, is the 1st FDA-approved neuromodulation technology for the treatment of HFrEF. In a meta-analysis, these investigators examined all randomized trial evidence to evaluate the effect of BAT on HF symptoms, QOL and NT-proBNP in patients with HFrEF. An individual patient data (IPD) meta-analysis was carried out on all eligible studies that randomized HFrEF patients to BAT + GDMT or GDMT alone (open label). Endpoints included 6-month changes 6MHW distance, Minnesota Living with Heart Failure (MLWHF) QOL score, NT-proBNP, and NYHA class in all patients and 3 subgroups. A total of 554 randomized patients were included. In all patients, BAT provided significant improvement in 6MHW distance of 49 m (95 % CI: 33 to 64), MLWHF QOL of -13 points (95 % CI: -17 to -10), and 3.4 higher odds of improving at least 1 NYHA class (95 % CI: 2.3 to 4.9) when comparing from baseline to 6 months. These improvements were similar, or better, in patients who had baseline NT-proBNP of less than 1,600 pg/ml, regardless of the CRT indication status. The authors concluded that an IPD meta-analysis suggested that BAT improved exercise capacity, NYHA class, and QOL in HFrEF patients receiving GDMT. These clinically meaningful improvements were consistent across the range of studies. BAT was also associated with an improvement in NT-proBNP in subjects with a lower baseline NT-proBNP.

These researchers stated that this IPD meta-analysis included only 2 randomized trials and with a limited number of patients restricting the ability to observe subtle differences in responses between patient cohorts of interest. Furthermore, both studies were open-label, which may result in bias in the more subjective endpoints. Nevertheless, the results were encouraging while they await longer term clinical outcomes from the 2nd post-market phase of the BeAT-HF Trial.

Babar and Giedrimiene (2022) noted that in the past 10 years, neuromodulation via BAT and VNS has emerged as an innovative approach for the treatment of HFrEF. These investigators carried out a review of the literature to examine the safety and effectiveness on neuromodulation for the treatment of HFrEF. Two independent researchers searched the PubMed, clinicaltrials.org, and the Cochrane databases for the most recent data on BAT and VNS published between 2013 and 2019; and a total of 9 studies were identified. BAT and VNS therapy consistently improved subjective HF parameters including NYHA functional class and MLWHF Questionnaire. Improvements in objective cardiac parameters such as left ventricular ejection fraction (LVEF) were less consistently observed; however, where present, ranged from +3 % to +6 %, in line with improvements observed after other guideline directed therapy such as left ventricular assist device (LVAD). Benefits of BAT showed a predilection for patients without CRT and effectiveness of VNS therapy varied with device type. The authors concluded that the clinical application of BAT and VNS was found to be limited due to low-powered data, inconsistencies in study design, short-term follow-up and lack of diversity in patient recruitment. These researchers stated that well-powered studies with consistent design, longer follow-up and diverse populations are needed before BAT and VNS can be incorporated into HF guidelines and clinical practice.

Ahmed et al (2024) stated that treatment strategies that modulate autonomic tone via interventional and device-based therapies have been studied as an adjunct to pharmacotherapy of HFrEF. In a meta-analysis, these investigators examined RCTs that assessed the effectiveness of device-based autonomic modulation for treatment of HFrEF. All RCTs testing autonomic neuromodulation device therapy in HFrEF were included in this analysis. Autonomic neuromodulation techniques included baroreflex activation (BRA), spinal cord stimulation (SCS), renal denervation (RD), and VNS. The pre-specified primary endpoints included mean change and 95 % CI of LVEF, NT-proBNP, and QOL measures including 6-minute hall walk distance (6-MHWD), and MLHFQ. NYHA functional class improvement was reported as odds ratios (ORs) and 95 % CI of improvement by at least 1 functional class. A total of 8 studies were identified that included 1,037 participants (2 BRA, 1 SCS, 3 RD, and 2 VNS trials). This included 6 open-label, 1 single-blind, and 1 sham-controlled, double-blind study. The mean age (± SD) was 61 (± 9.3) years. The mean follow-up time was 7.9 months; 20 % of the total patients were women, and the mean BMI (± SD) was 29.86 (± 4.12). Autonomic neuromodulation device therapy showed a statistically significant improvement in LVEF (4.02 %; 95 % CI: 0.24 to 7.79), NT-proBNP (-219.80 pg/ml; 95 % CI: -386.56 to -53.03), NYHA functional class (OR 2.32; 95 % CI: 1.76 to 3.07), 6-MHWD (48.39 m; 95 % CI: 35.49 to 61.30), and MLHFQ (-12.20; 95 % CI: -19.24 to -5.16) compared to control. The authors concluded that in patients with HFrEF, the use of autonomic neuromodulation device therapy was associated with improvement in LVEF, reduction in NT-proBNP, and improvement in patient-centered QOL outcomes in mostly small open-label trials. Moreover, these researchers stated that large, double-blind, RCTs, or sham-controlled trials designed to detect differences in hard cardiovascular outcomes are needed before widespread use and adoption of autonomic neuromodulation device therapies in HFrEF.

In a prospective, randomized, 2-arm, parallel-group, open-label, non-implanted control, multi-center trial (the BeAT-HF Trial), Zile et al (2024) tested the hypothesis that treatment with BAT would significantly reduce cardiovascular mortality and HF morbidity and provide long-term safety and sustainable symptomatic improvement. Participants were NYHA class-III patients with EF of 35 % or less, previous HF hospitalization or NT-proBNP of greater than 400 pg/ml, no class-I indication for CRT and NT-proBNP of less than 1,600 pg/ml. They were randomized to BAT plus optimal medical management (BAT group) or optimal medical management alone (control). The primary end-point was cardiovascular mortality and HF morbidity; additional pre-specified end-points included durability of safety, QOL, exercise capacity (6MHWD), functional status (NYHA class), hierarchical composite win ratio, freedom from all-cause death, LVAD implantation, and heart transplant. Overall, 323 patients had 332 primary events, median follow-up was 3.6 years/patient. Both primary end-point (rate ratio 0.94, 95 % CI: 0.57 to 1.57; p = 0.82) and components of the primary end-points were not significantly different between BAT and control. The system- and procedure-related major adverse neurological and cardiovascular event-free rate remained 97 % throughout the trial. Symptom improvement (QOL, 6MHWD, NYHA class, all nominal p < 0.001) in the BAT group was durable in time, sustainable in extent. Win ratio (1.26, 95 % CI: 1.02 to 1.58) and freedom from all-cause death, LVAD implantation, heart transplant (HR 0.66, 95 % CI: 0.43 to 1.01) favored the BAT group but did not reach statistical significance. The authors concluded that baroreflex activation therapy did not result in a significant difference in the composite primary end-point, CV mortality and HF morbidity, or the individual components of the primary end-points compared with control. However, both long-term measures of safety and symptomatic improvement favored the BAT group, were durable over time, and sustainable in the extent of their effects.

The authors stated that this trial had several drawbacks. First, the BeAT-HF Trial was an unblinded study with a non-implanted control. This design decision constituted one important limitation, especially as it relates to a differential placebo effect in the control versus BAT group. Based on data provided in this trial, these researchers could not rule out the presence of a persistent placebo effect (in the BAT device group) that could have contributed to the differences observed between the control and BAT groups in the patient-centered symptomatic end-points. These investigators stated that future studies with BAT are planned, and will employ a double-blind, implanted control design that will address this limitation. Second, the sample size in the BeAT-HF Trial was likely to be under-powered for at least the CV mortality end-point of the composite end-point. It was for this and other reasons that freedom from all-cause death, LVAD and heart transplant was also examined. Third, the entrance criteria (especially NT-proBNP of less than 1,600 pg/ml) may have limited the expected number of mortal and morbid events. Fourth, the BeAT-HF Trial focused exclusively on patients with HFrEF. Given the BAT mechanism of action and given the fact that autonomic dysfunction is present in all patients with chronic HF, there is every reason to anticipate that BAT would have salutary effects on patients with HF and mildly reduced and preserved EF. These researchers noted that these studies are currently being planned.

Sears et al (2024) discussed the long-term QOL outcomes associated with BAT in patients with HFrEF. The study builds on findings from the BeAT-HF Trial, which examined the effectiveness of BAT compared to guideline-directed medical therapy (GDMT) alone. The BeAT-HF Trial consisted of 2 phases: a pre-market phase that examined the effectiveness of BAT after 6 months in 264 subjects, and a post-market phase that extended follow-up to assess long-term outcomes in 323 subjects. The pre-market phase showed significant improvements in various metrics, including the 6MWD, MLWHF scores, NYHA functional class, and NT-proBNP levels. These findings supported FDA approval for BAT in specific patient populations. The current study aimed to report changes in both HF-specific and general QOL parameters at 24 months in patients receiving BAT plus GDMT compared to those receiving GDMT alone. The study was carried out in compliance with ethical standards, and informed consent was obtained from all participants. A total of 323 subjects with HFrEF were randomized into 2 groups: one receiving BAT plus GDMT (BAT arm) and the other receiving GDMT alone (control arm). QOL was assessed at baseline and 24 months using the MLWHF and the EuroQol 5-Dimensions Long (EQ-5D) tool. The MLWHF questionnaire assesses both physical and emotional dimensions, while the EQ-5D assesses overall health status and index score. At the 24-month follow-up, 124 subjects from the BAT arm and 105 from the control arm completed the assessment. Changes in QOL scores were compared between the 2 groups using a generalized estimating equation repeated measurement model. The results indicated that the BAT arm experienced significant improvements in both HF-specific and general QOL measures compared to the control arm. Specifically, the BAT group showed: A reduction in the MLWHF physical score (Δ = -3.7; 95 % CI: -6.3 to -1.1) and emotional score (Δ = -2.0; 95 % CI: -3.5 to -0.6). Higher scores on the EQ-5D index and overall health status compared to the control group. The study highlighted that the most responsive areas to BAT included the ability to return to usual activities and reduced pain/discomfort.

The authors stated that this study had several drawbacks. First, although no known selection bias was reported, the study's design could not completely rule out "survivor bias" due to approximately 30 % attrition over the 2-year follow-up. Second, the lack of blinding in this study may have resulted in expectancy effects, where participants' expectations of improvement influenced their reporting of outcomes. The impact of these effects on long-term QOL measures remains uncertain. Third, while the sample size was adequate for statistical analysis, these investigators suggested that larger studies could provide more robust insights into specific patient experience variables. They emphasized the importance of further research to optimize QOL in HF patients using device-based therapies like BAT. These researchers stated that while this trial provided valuable insights into the long-term benefits of BAT, further investigations are needed to examine the mechanisms behind these improvements and to assess the impact of expectancy effects in similar interventions.

Chow et al (2024) noted that baroreflex activation therapy (BAT) is an emerging device-based treatment for patients with heart failure with a reduced ejection fraction (HFrEF) refractory to maximally tolerated goal-directed medical therapy. Currently, there is sparse literature on the critical role that vascular surgeons serve in the delivery of this novel therapy. In a single-center study, these researchers described the creation of a BAT program and elaborated on the function of vascular surgeons in the multi-disciplinary HF team. The pre-operative evaluation, peri-operative care, and post-operative course of patients receiving BAT from March 2022 to August 2023 were retrospectively analyzed. A total of 11 patients were evaluated by a dedicated HF cardiologist for medical eligibility and assessed by a vascular surgeon for technical feasibility. Of the 11 patients, 7 were men (63.6 %); the median age was 60.5 years (range of 44 to 73). No patient (0.00 %) had existing carotid artery disease (CAD) and 1 patient (9.1 %) had undergone previous neck radiation therapy. All 11 patients (100 %) had an existing cardiac implantable electronic device, and BAT implantation was carried out on the same side as the cardiac implantable electronic device in 2 patients (18.1 %). A total of 4 patients (36.4 %) required pre-operative hospital admission for medical optimization before surgery. The median length of surgery was 82 mins (range of 58 to 113), and the median length of stay (LOS) in the hospital after surgery was 1 day (range of 0 to 6 days). No major adverse neurologic or cardiovascular events, cranial nerve injuries, device complications requiring re-intervention, or HF-related mortality at 6 months occurred; and 3 patients (27.3 %) experienced extraneous stimulations, which affected BAT tolerability. Within 6 months after BAT implantation, no significant improvements were observed for several HF disease burden markers compared with 6 months before BAT implantation. The authors concluded that these early findings showed that BAT implantation was a safe procedure with rare complications. These investigators stated that vascular surgeons play an important role in the multi-disciplinary delivery of this novel device-based therapy. They stated that more data are needed to examine if BAT is beneficial in the treatment of HF with a reduced ejection fraction refractory to maximally tolerated goal-directed medical therapy.

Schafer et al (2024) stated that arterial hypertension (aHTN) plays a fundamental role in the pathogenesis and prognosis of HFpEF. The risk of HF increases with therapy-resistant arterial hypertension (trHTN), defined as inadequate blood pressure (BP) control ≥ 140/90 mmHg despite taking ≥ 3 anti-hypertensive medications including a diuretic. In an observational, single-center study, these investigators examined the effects of the BP-lowering BAT on cardiac function and morphology in patients with trHTN with and without HFpEF. A total of 64 consecutive patients who had been diagnosed with trHTN and received BAT implantation between 2012 and 2016 were prospectively observed. Office BP, electrocardiographic and echocardiographic data were collected before and after BAT implantation. Mean patients' age was 59.1 years, 46.9 % were men, and mean BMI was 33.2 kg/m2. The prevalence of diabetes mellitus (DM) was 38.8 %, AF was 12.2 %, and chronic kidney disease (CKD) stage 3 or higher was 40.8 %; 28 patients had trHTN with HFpEF, and 21 patients had trHTN without HFpEF. Patients with HFpEF were significantly older (64.7 years versus 51.6 years, $p < 0.0001$), had a lower BMI (30.0 versus 37.2 kg/m2, $p < 0.0001$), and suffered more often from CKD-stage 3 or higher (64 % versus 20 %, $p = 0.0032$). After BAT implantation, mean office BP dropped in patients with and without HFpEF (from 169 ± 5/86 ± 4 to 143 ± 4/77 ± 3 mmHg [p = 0.0019 for SBP and 0.0403 for DBP] and from 170 ± 5/95 ± 4 to 149 ± 6/88 ± 5 mmHg [p = 0.0019 for SBP and 0.0763 for DBP]), while a significant reduction of the intake of calcium-antagonists, α2-agonists and direct vasodilators, as well as a decrease in average dosage of ACE-inhibitors and α2-agonists could be observed. Within the study population, a decrease in HR from 74 ± 2 to 67 ± 2 min-1 ($p = 0.0062$) and lengthening of QRS-time from 96 ± 3 to 106 ± 4 ms ($p = 0.0027$) and QTc-duration from 422 ± 5 to 432 ± 5 ms ($p = 0.0184$) were detectable. The PQ duration was virtually unchanged. In patients without HF, no significant changes of echocardiographic parameters could be observed. In patients with HFpEF, posterior wall diameter decreased significantly from 14.0 ± 0.5 to 12.7 ± 0.3 mm ($p = 0.0125$), left ventricular mass (LVM) declined from 278.1 ± 15.8 to 243.9 ± 13.4 g ($p = 0.0203$), and e' lateral increased from 8.2 ± 0.4 to 9.0 ± 0.4 cm/s ($p = 0.0471$). The authors concluded that BAT reduced SBP and DBP; and was associated with morphological and functional improvement of HFpEF.

Moreover, these researchers stated that further research and larger-scale studies are needed to confirm these findings and to examine if the particular effect of BAT on diastolic function in patients with HFpEF can be verified.

The authors noted that being a single-center evaluation in a rare indication, the study was limited to a small sample size with missing data. Moreover, due to its observational nature, it lacked a control group. Electrocardiograms were available for only 33 patients. As ECG data were derived from single ECG measurements at baseline and follow-up, HR variability during the day may result in incorrect data. The Devereux formula was employed to calculate LVM. This formula is based on linear measurements of ventricular wall and internal diameters without taking into account the 3D ventricular structure, asymmetric changes or interindividual differences in LV morphology. Furthermore, the formula might under-estimate LV hypertrophy; thus, MRI would have allowed a better evaluation of LVM. Nevertheless, estimation of LVM by Devereux formula is an accepted tool, and a possible under-estimation of LVM in this trial affected both baseline and follow-up data; therefore, the change of LVM should be valid. It would have been of interest to correlate cardiac morphological changes with clinical outcome. Unfortunately, no data on clinical endpoints were available.

Falco et al (2024) stated that HF is a complex and progressive disease marked by substantial morbidity and mortality rates, frequent episodes of decompensation, and a reduced QOL (QoL), with severe financial burden on healthcare systems. In recent years, several large RCTs have widely expanded the therapeutic armamentarium, underlining additional benefits and the feasibility of rapid titration regimens. This notwithstanding, mortality is not declining, and hospitalizations are constantly increasing. It is widely acknowledged that even with guideline-directed medical therapy (GDMT) on board, HF patients have a prohibitive residual risk, which highlights the need for innovative therapeutic options. In this scenario, ground-breaking devices targeting valvular, structural, and autonomic abnormalities have become important tools in HF management. This has resulted in a full-fledged translational boost with several novel devices in development. These investigators provided an updated review on both approved and investigated devices. They noted that a specific study of HFpEF patients with resistant hypertension is ongoing (BAROSTIM THERAPY In Heart Failure With Preserved Ejection Fraction). Moreover, these researchers stated that although it is reasonable to employ devices in patients who cannot attain GDMT, there is an urgent need for studies specifically focused on investigating the order and fast escalation of all HF treatments, including medications and devices. This is necessary even for individuals who are considered to have a high tolerance for medications, as the residual risk remains significant.

Estep et al (2024) stated that heart failure (HF) is one of the major challenges of our time, given its increase in prevalence and related mortality rates. Foundational pharmacotherapies, including angiotensin receptor neprilysin inhibitors (ARNIs), beta-blockers, mineralocorticoid receptor antagonists (MRAs), as well as sodium-glucose co-transporter inhibitors (SGLTis), have been established for HF with reduced ejection fraction (HFrEF). Moreover, recent studies have established the role of SGLTis in patients with HF with preserved ejection fraction (HFpEF). However, even with these therapies, a substantial residual risk persists in both HFrEF and HFpEF. Alongside pharmacological advancements, device-based therapies have shown effectiveness in HF management, including implantable cardioverter-defibrillators (ICDs) and cardiac resynchronization therapy (CRT). More recently, devices such as cardiac contractility modulation (CCM) and baroreflex activation therapy (BAT) have been approved by the FDA, although they lack comprehensive guideline recommendations. These investigators outlined the unmet needs in chronic HF, reviewed contemporary data and provided a framework for integrating novel device-based therapies into current clinical work-flows. They emphasized the importance of early diagnosis and phenotyping, proper patient stratification, and a personalized approach to combining pharmacological and device therapies. The authors highlighted the need for further research into device interactions and patient selection to optimize outcomes, while recognizing the need for a more integrated approach to treatment so as to address the unmet needs and residual risks in HF management.

Bustos-Merlo et al (2025) stated that the number of patients suffering from refractory hypertension and advanced-stage chronic HF (CHF) is progressively increasing. In recent years, device-mediated therapies have been developed as an alternative or adjunct to conventional medical treatment. These investigators described the clinical experience in a series of patients with refractory hypertension following the implantation of BAT. They analyzed 5 patients with refractory hypertension, 1 also had CHF, treated in a specialized cardiovascular risk clinic. After the implantation of the Barostim device with an average activation of 3.64mA (ranging from 2.80 to 5.4), there was a mean reduction of 30 ± 7.68 mmHg ($p = 0.001$) and 13.40 ± 9.07 mmHg ($p = 0.03$) in systolic and diastolic BP (SBP and DBP), respectively, as measured by ambulatory BP monitoring (ABPM), along with a heart rate (HR) reduction of 25 ± 9.13 bpm ($p = 0.004$). A reduction in the number of anti-hypertensive medications required for BP control was observed, with an average of 5.2 medications, as well as an improvement in functional class. No adverse events (AEs) were recorded in this patient series. The authors concluded that currently, BAT is considered a compassionate-use alternative for BP control in patients with refractory hypertension and failure of pharmacotherapies and other invasive techniques, such as renal denervation. These researchers stated that current clinical guidelines do not view it as a therapeutic option, because of the low level of evidence (IIIc).

The authors stated that the drawbacks to this study were a consequence of its small and heterogeneous sample. Meaningful extrapolations could not be drawn from these data. The series could be under-powered to detect low frequency, but clinically significant, procedure- and device-related complications. This trial only served as hypothesis-generating and could not definitively show that BAT implantation is technically achievable and associated with minimal complications. These researchers stated that long-term data and prospective, randomized, controlled, and blinded studies are needed to prove that BAT is a safe and effective therapy for patients with HFrEF refractory to maximally tolerated GDMT. However, much of the literature describing BAT is from Europe, where the technology has been in use for a longer period and there are a wider range of clinical applications, including resistant hypertension. Data germane to the U.S. population is still being collected but could be eclipsed by the rate at which new technologies are being developed, such as endovascular options, which are on the horizon. These novel data represent the real-world experience of the most up-to-date BAT technology from the perspective of vascular surgeons treating patients with HFrEF in the United States.

Abraham et al (2025) defined the association between BAT and subsequent survival and hospitalization. De-identified data from the Premier Healthcare Database (PHD) were employed for this analysis. The PHD is an all payer data-base inclusive of inpatient and outpatient encounters from over 1,300 healthcare facilities. Items billable to a patient or health insurance provider -- including hospital services, medical devices, and drugs -- were captured in the chargemaster. The study population was defined using CPT code 0266T, HCPCS code C1825, or “Barostim” in the chargemaster description from January 1, 2016 through June 30, 2023. Patient demographics, medical history, healthcare utilization, and mortality were assessed. Healthcare utilization was defined using hospitalization and emergency department (ED) encounters, classified as all-cause, cardiovascular (CV), non-CV, and HF-related. Rates of overall hospital visits (hospitalization and ED encounters) and rates of hospitalization were evaluated up to 12 months pre-BAT implant and for the entire post-BAT implant follow-up. The post-implant duration was defined as the time between BAT implant and the data extraction date of June 30, 2023 or death. Hospitalization length of stay (LOS) was compared between 12 months pre-implant and 12 +/- 3 months post-implant. All-cause and CV hospitalizations were defined according to the Heart Failure Academic Research Consortium. HF hospitalizations and ED encounters were captured through the corresponding Diagnosis Related Group and ICD-10 codes, respectively. Non-CV encounters were defined as any encounter that did not originate from a cardiac or vascular cause. From 307 million unique patients in the PHD, a total of 4 306 patients underwent BAT implantation and comprise the study cohort. As Barostim was approved in 2019, the majority of implants (275/306; 90 %) and follow-up occurred in 2021 to 2023. Mean age was 66 +/- 12 years, 79 (25.8 %) were women, and 65 (21.2 %) were Black. Coronary artery disease (CAD), prior myocardial infarction (MI), and type 2 diabetes mellitus (T2DM) were common. Most patients (52.9 %) had an implantable cardioverter defibrillator/cardiac resynchronization therapy defibrillator (ICD/CRT-D), and a cardiac resynchronization therapy pacemaker (CRT-P) was present in 18.3 % of patients. The post-implant follow-up duration was 1.92 +/- 1.87 years, representing a total of 586 patient-years of follow-up. The mortality rate at 1 year and 2 years post-implant was 6.3 % and 12.3 %, respectively. Pre- and post-implant incidence rates for all-cause, CV, non-CV, and HF hospitalizations and hospital visits were shown in Figure 1. Compared with the 12-month period before BAT implant, there were significantly lower post-implant rates of hospitalizations (all-cause: 85 % rate reduction; CV: 81 % rate reduction; non-CV: 88 % rate reduction; HF: 71 % rate reduction; $p < 0.001$ for all categories) and hospital visits (all-cause: 86 % rate reduction; CV 84 % rate reduction; non-CV: 87 % rate reduction; HF: 85 % rate reduction; $p < 0.0001$ for all categories). As shown in Figure 2, there were significant reductions in hospitalization LOS in the 12 +/- 3 months post-implant versus the 12 months pre-implant. The authors concluded that the findings from this study supported the hypothesis that, in a real-world national administrative data-base, implantation of BAT was associated with reductions in all-cause, CV, non-CV, and HF hospital visits and LOS while accounting for the pandemic's effect on the decrease of healthcare utilization. Broader and specialized disease management changes associated with BAT as well as COVID-19 pandemic related changes in healthcare utilization likely influenced the decreased trends in hospital visits post-BAT as well. Moreover, these researchers stated that prospective clinical trials or larger real-world data-sets that include concomitant control subjects will provide more accurate assessment of BAT effectiveness in reducing healthcare utilization.

The authors stated that this trial had several drawbacks. First, chargemaster or claims data-base studies examining event rates before and after an intervention are inherently limited by several factors, including the lack of a concomitant control group, inability to fully characterize HF severity in the study population, as well as inability to examine if disease management differed over time. Such studies are not intended to accurately identify a point-estimate of effectiveness but are useful in describing real-world experience and consistency of results compared with prospective RCTs. Second, the pandemic's impact on decreased healthcare resource utilization represents a limitation, as it may have influenced the observed trends in this analysis. It is always possible that advancements and adoption of novel medical therapies, such as SGLT2-inhibitors, or higher prescription rates of other guideline-directed medical therapy may contribute to improved outcomes independent of BAT. Third, more sophisticated disease attributed to specialized HF centers may have started after BAT implantation, including close follow-up in the first 3 months as BAT was up-titrated, which potentially reduced risk for post-implant hospitalizations.

Furthermore, UpToDate reviews on "Overview of the management of heart failure with reduced ejection fraction in adults" (Colucci, 2025) and "Treatment and prognosis of heart failure with preserved ejection fraction" (Borlaug and Colucci, 2025) do not mention the Barostim as a management option.

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The above policy is based on the following references:

Drug-Resistant Hypertension

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Other Indications

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